ICTP-SAIFR EGSnrc Monte Carlo School for Medical Physics
IFT-UNESP, São Paulo, Brazil
March 30 – April 3, 2026

LECTURE 11 · EGSnrc course

Phase space file analysis with beamdp

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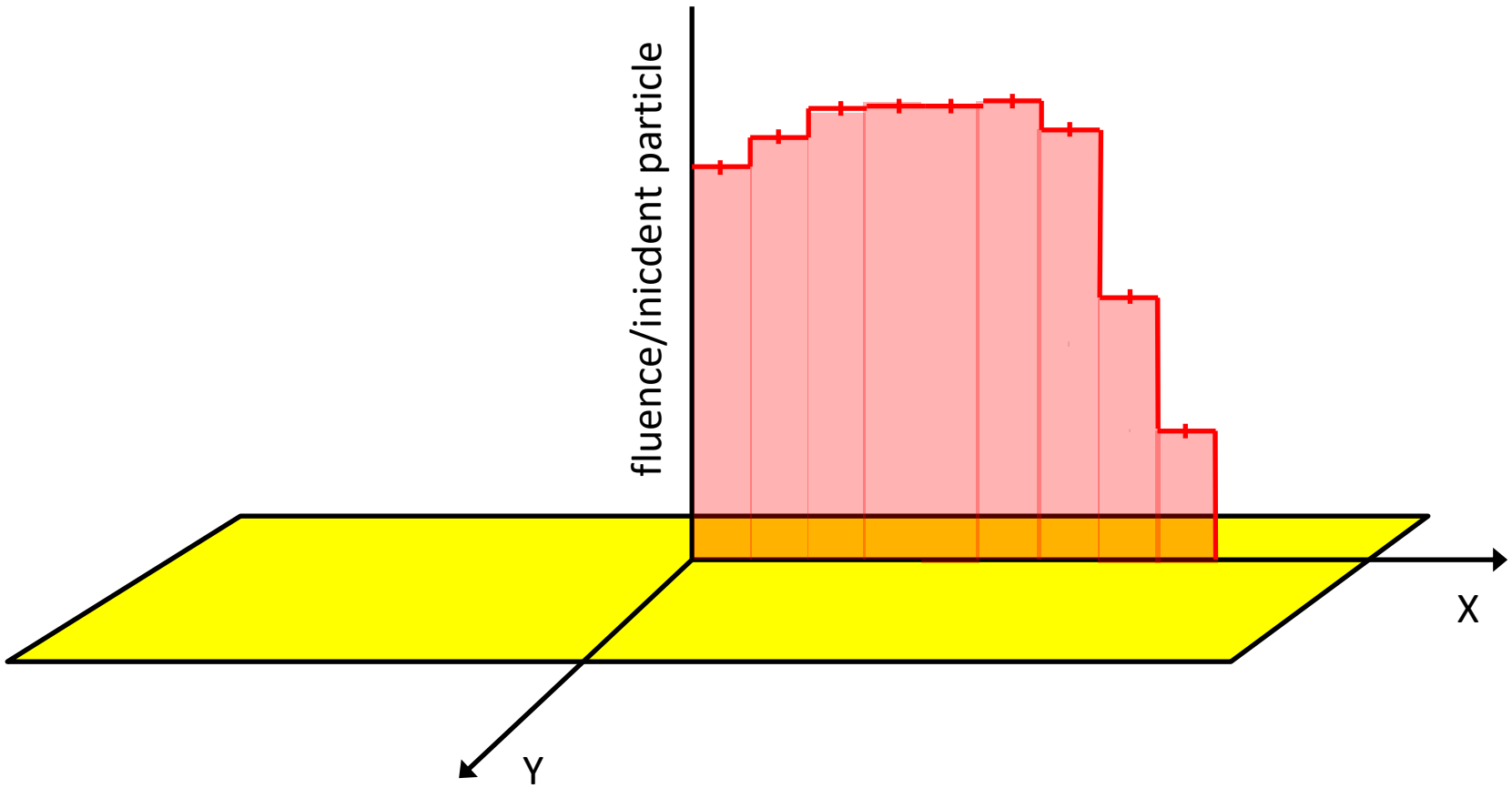
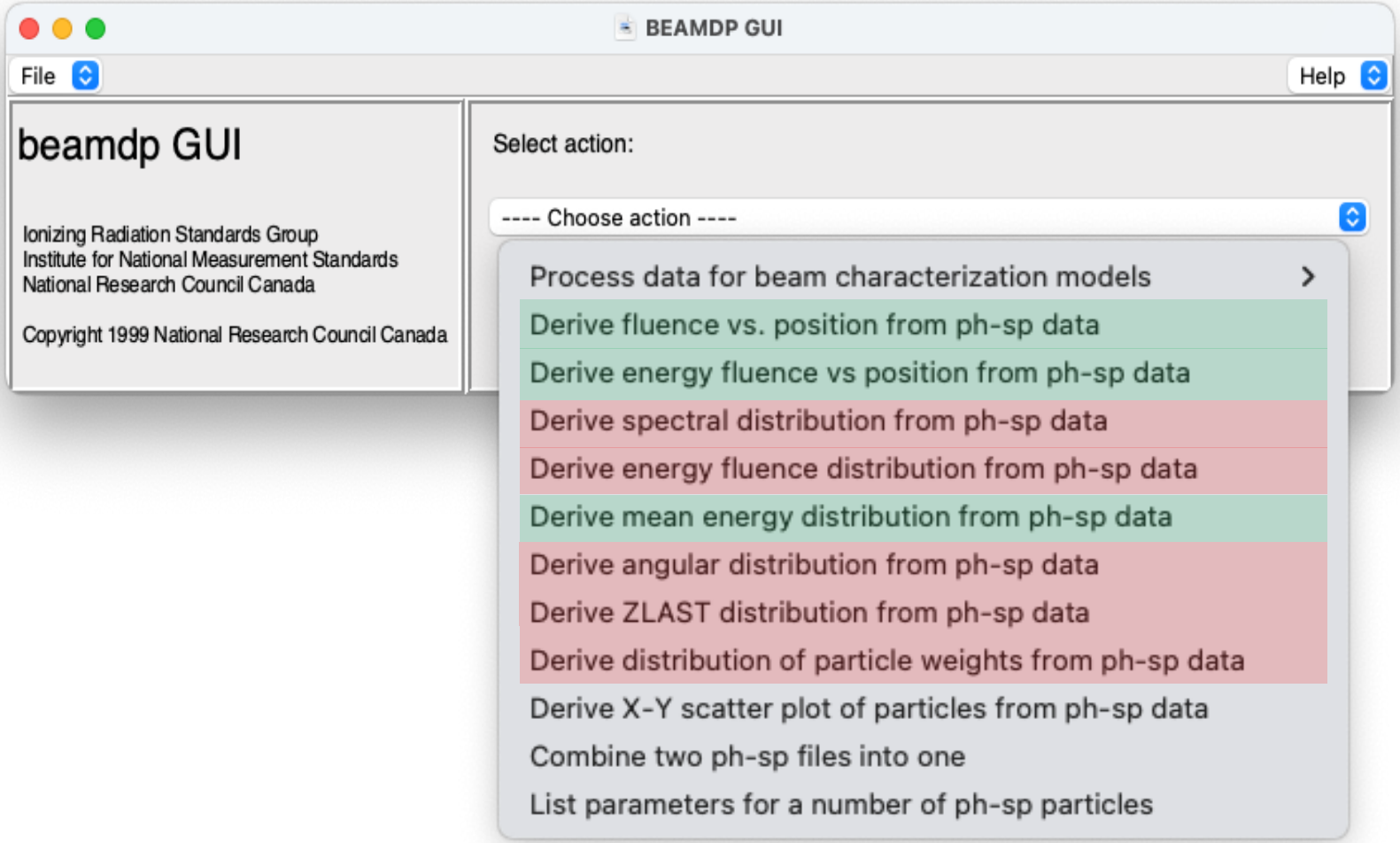
beamdp analyzes data stored in (2D) phase space files

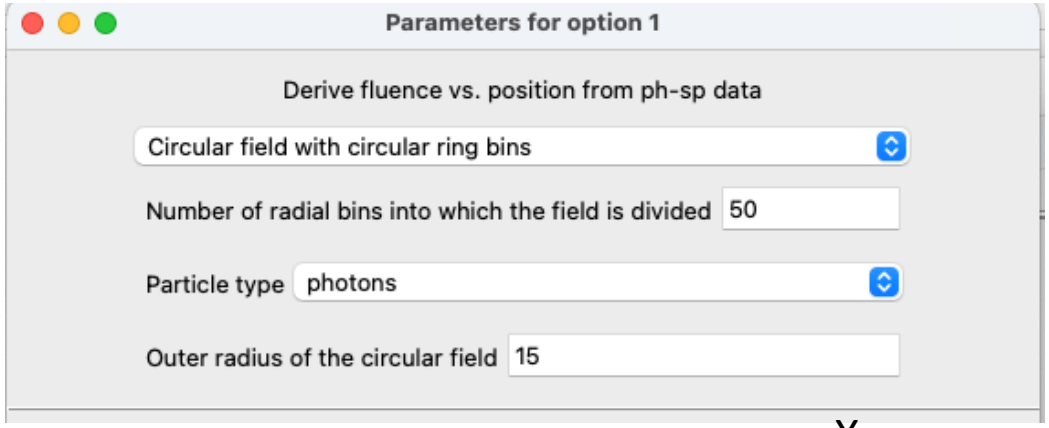


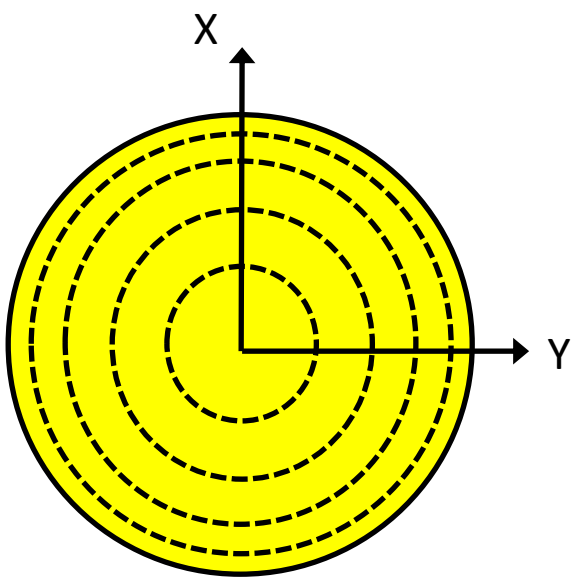
Spatial vs Spectral quantities

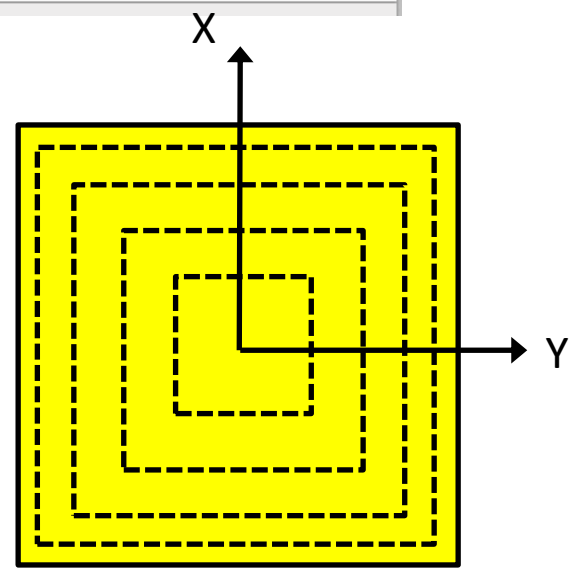


Spatial fields

- circular or square fields centered at origin divided into bins of equal area

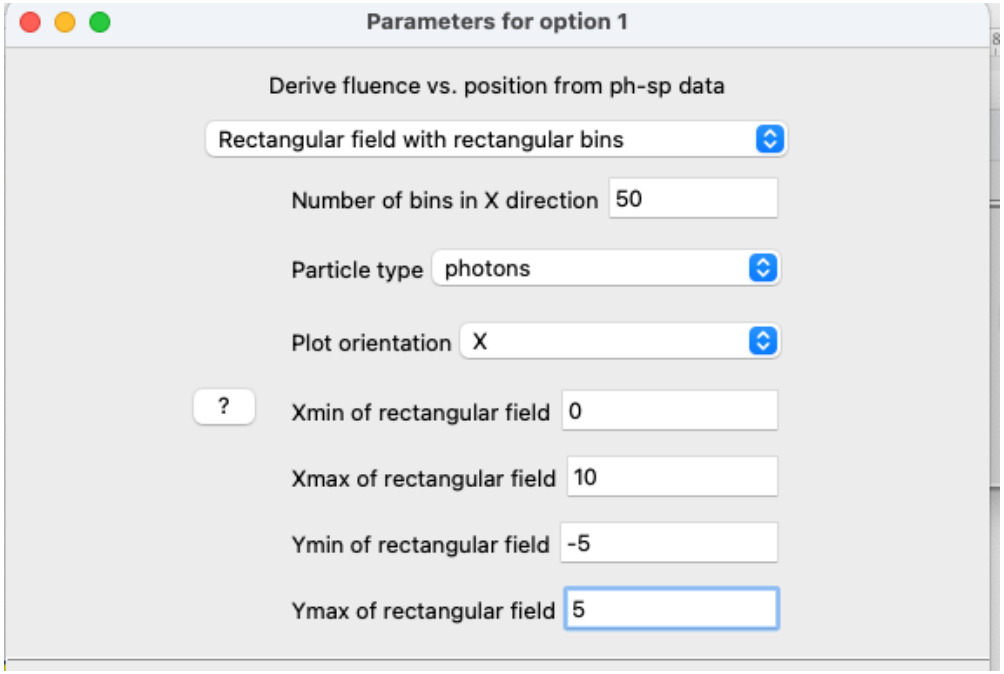


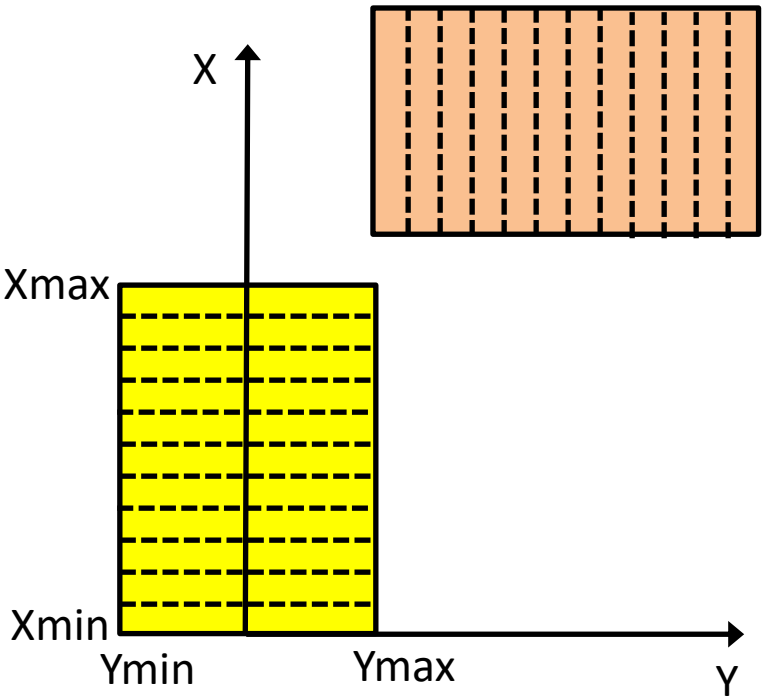




Spatial fields

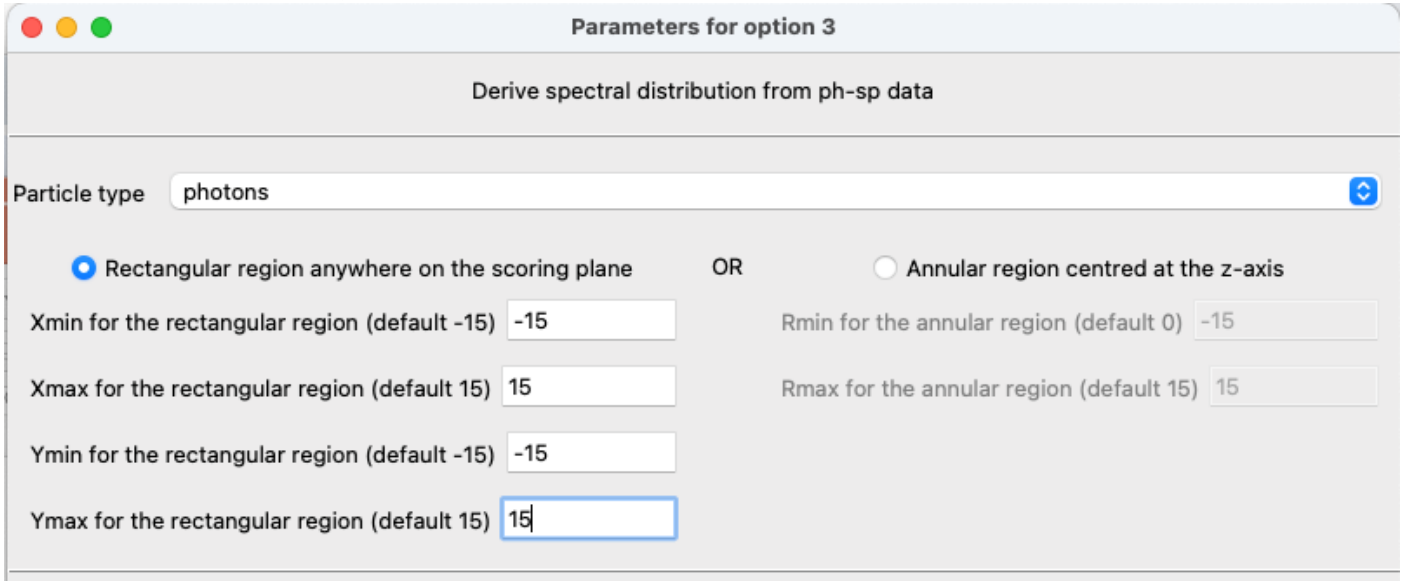
- rectangular fields anywhere on the phase space plane divided into bins of equal width
- bins (plots) oriented in X- or Y-direction

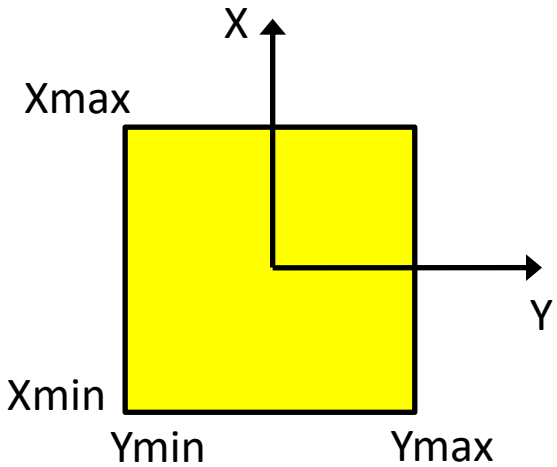


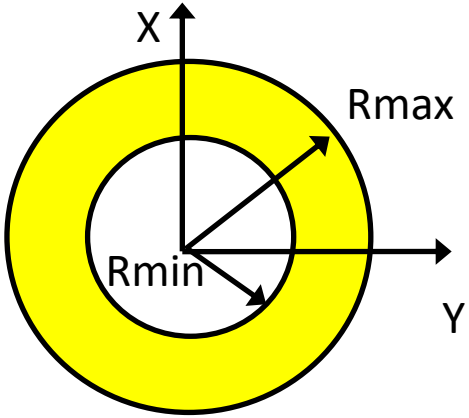


Spectral fields

- Rectangle anywhere on the plane or annular region centered at origin

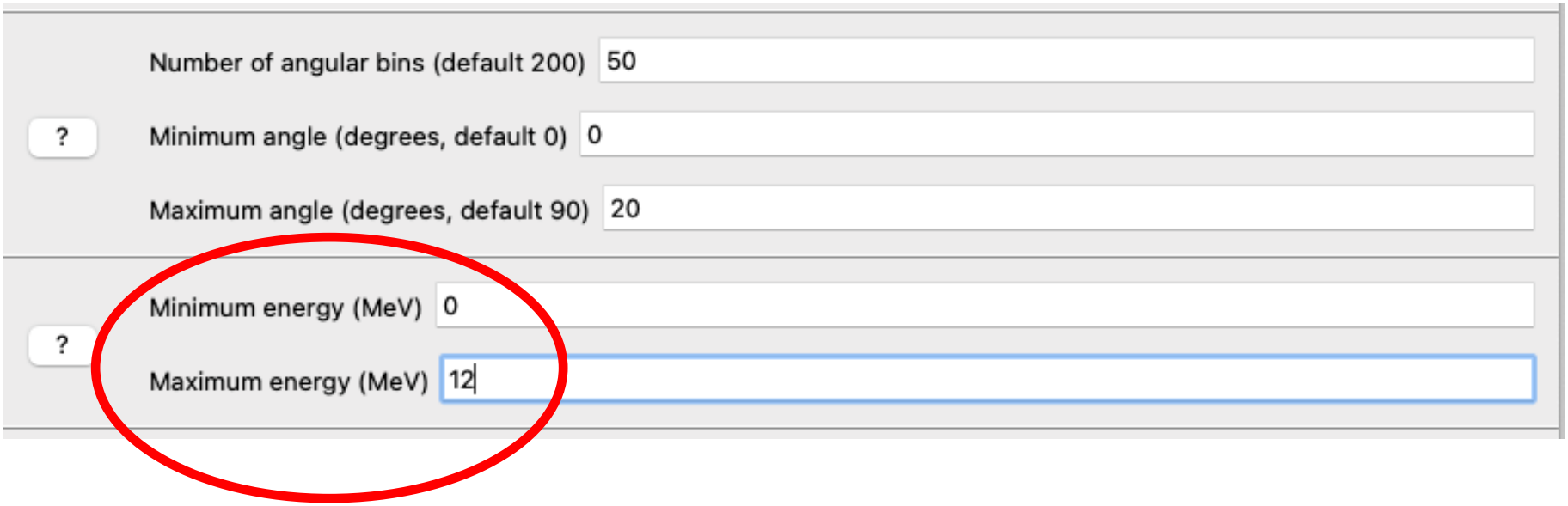






Spectral quantities

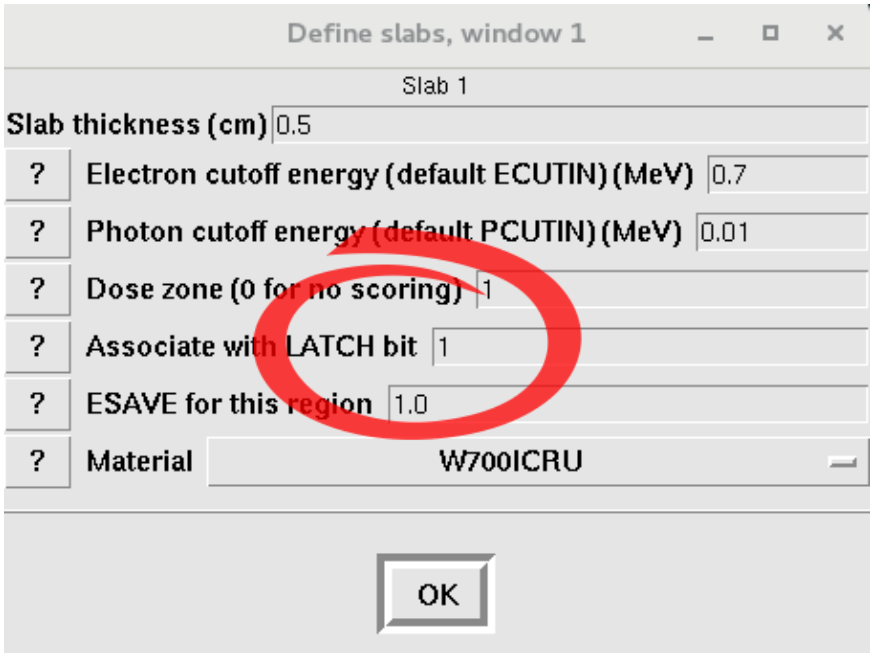
- Specify range and number of bins
- e.g. Angular distribution:

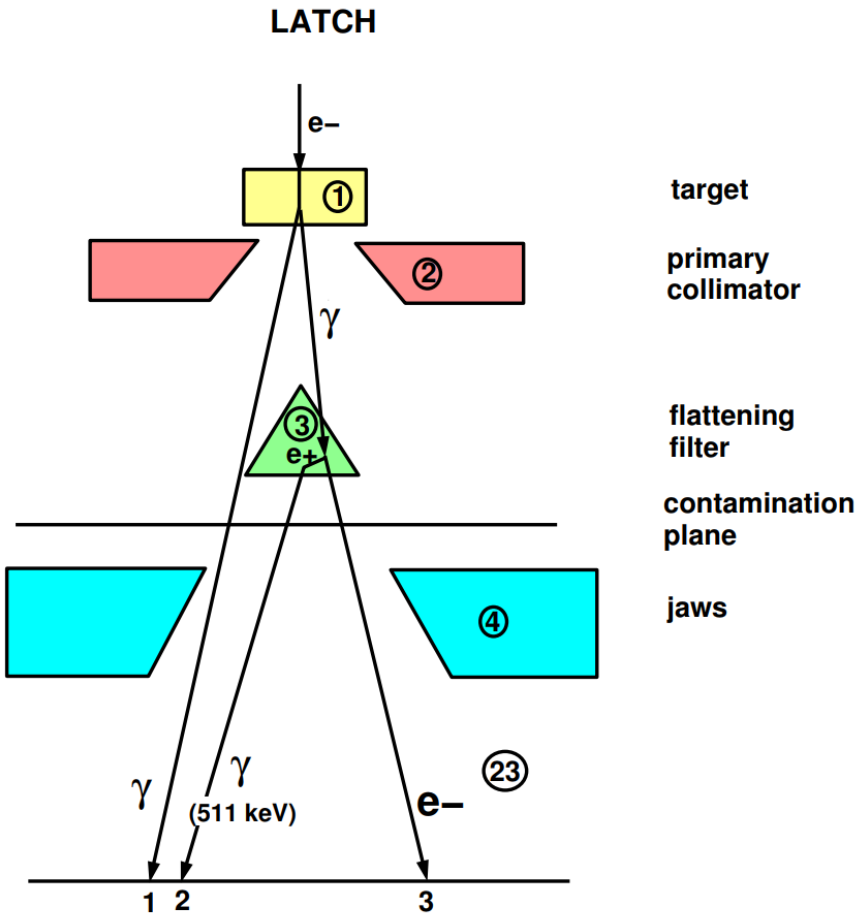


For angular distribution only: also specify an energy range

LATCH Review I

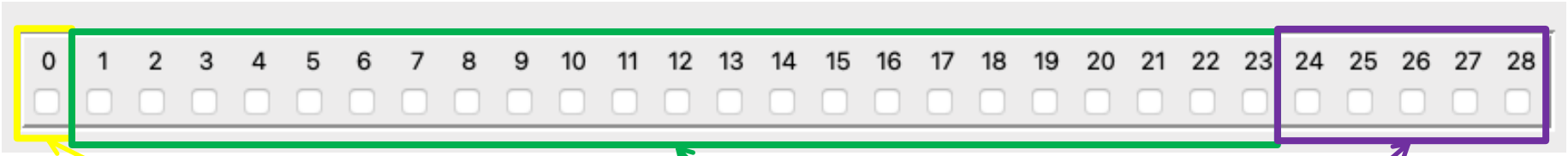
From beamnrc_gui:





- `LATCH` is a 32-bit integer variable that travels with the particle
- `LATCH` bits are associated with regions in BEAMnrc or other application

LATCH Review II



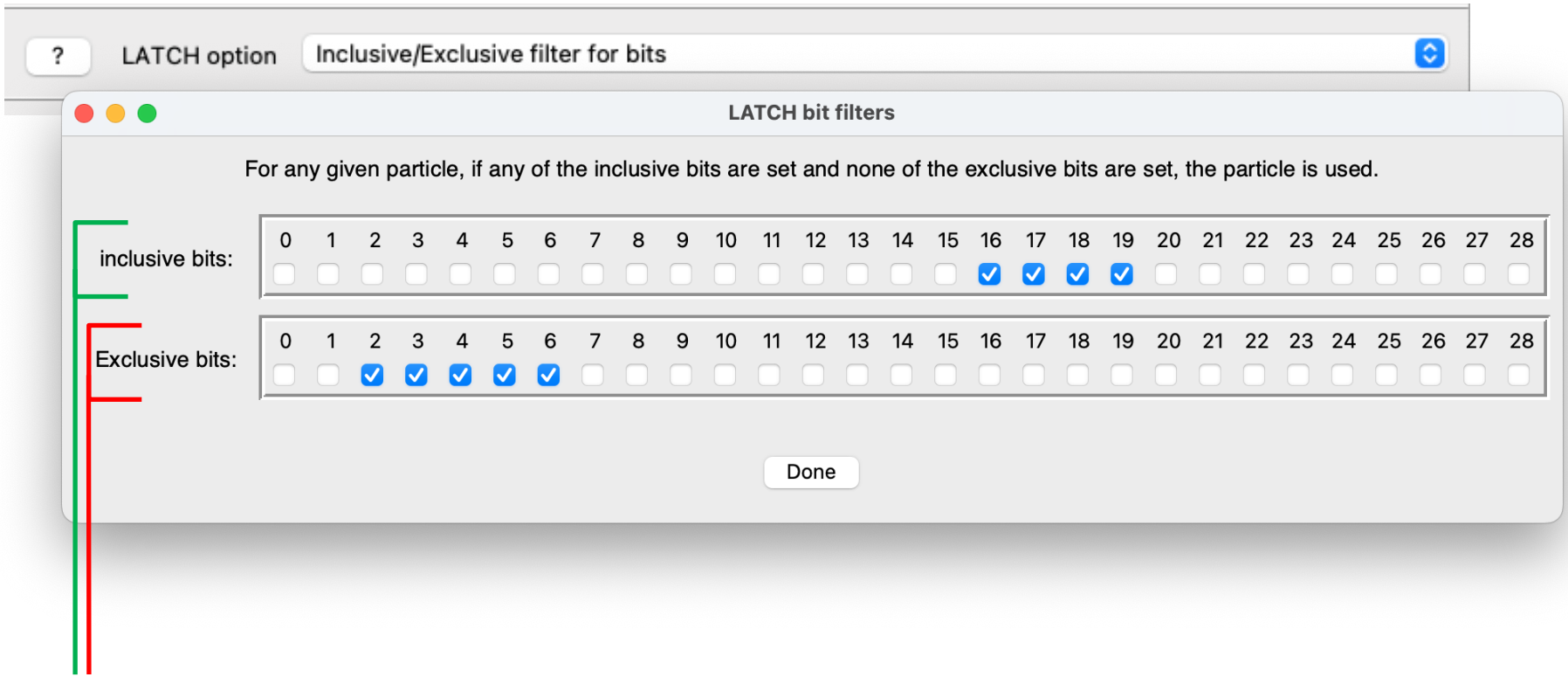
Set if brems or positron annihiliation occurred

Set if particle interacted/passed through corresponding region

- **For secondaries: 24-28, downshifted by 24 bits, store the bit (1-23) associated with the region where they were created**
- **For primaries: not set**

- **In BEAMnrc, AIR regions default to bit 23**

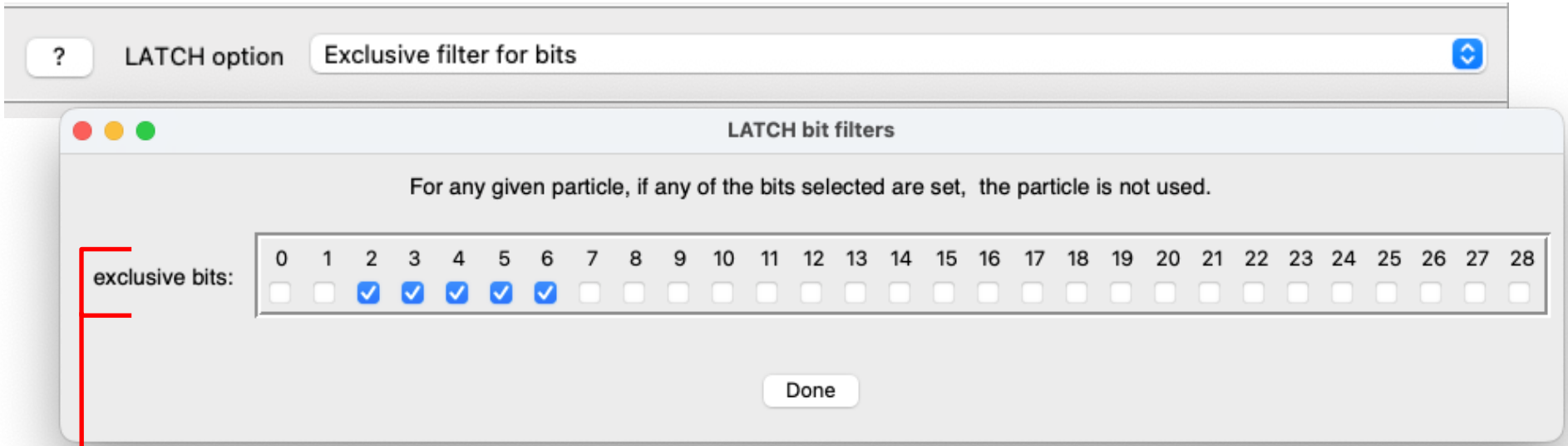
BEAMDP LATCH filtering: inclusive/exclusive for bits



Include the particle only if any "inclusive bits" are set

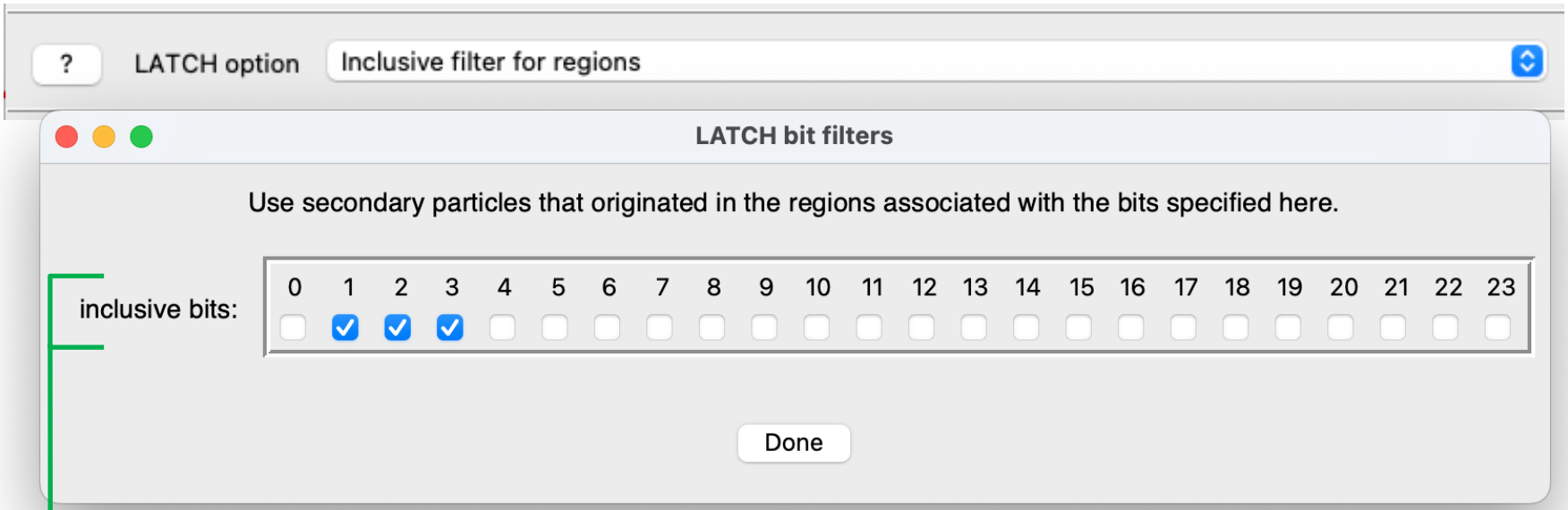
AND none of the "Exclusive bits" are set

Latch filtering: exclusive for bits



Include the particle only if none of the "exclusive bits" are set

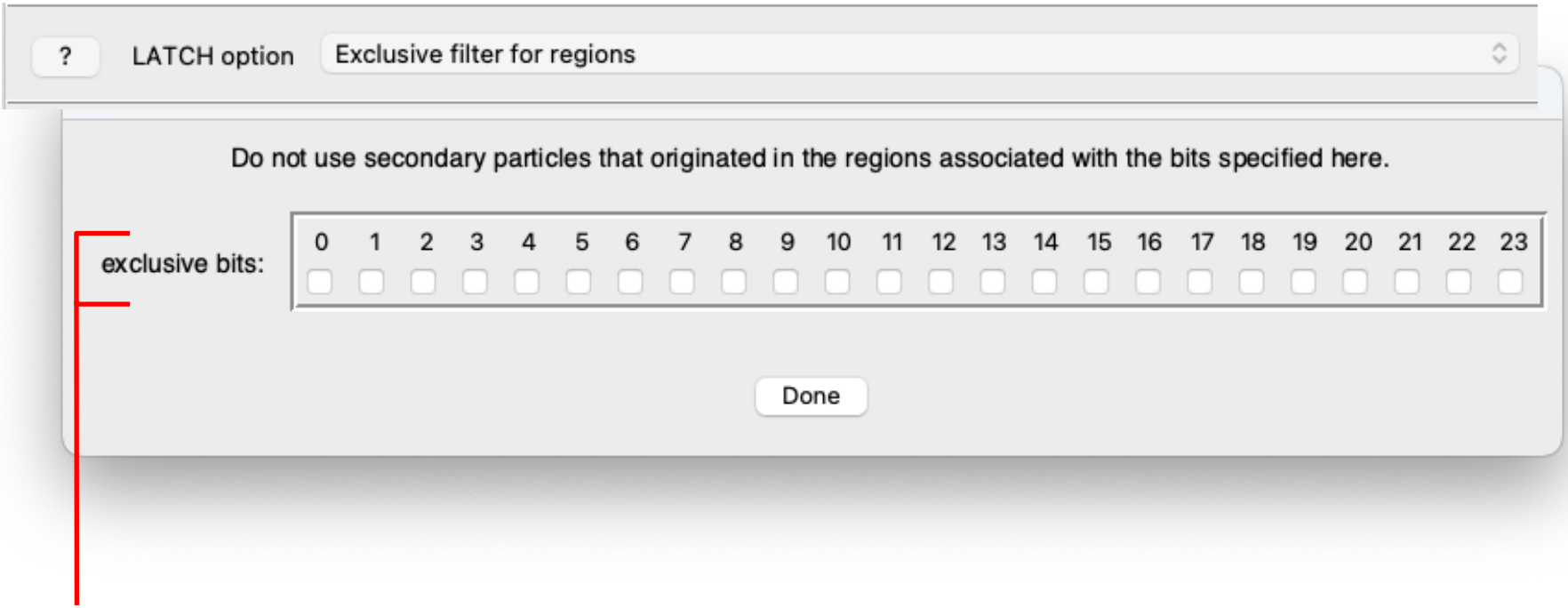
Latch filtering: inclusive for regions



Include only secondary particles originating in regions associated with "inclusive bits"

- **`LATCH` bits associted with regions in BEAMnrc or other application**
- **Stored in bits 24-28 of `LATCH`**
- **Region 23 is usually reserved for air**

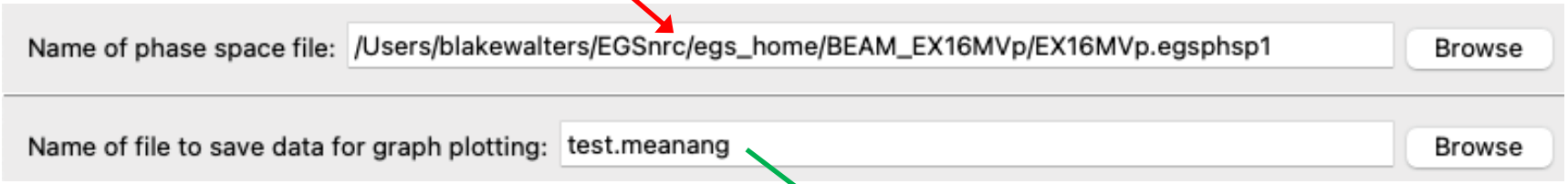
Latch filtering: exclusive for regions



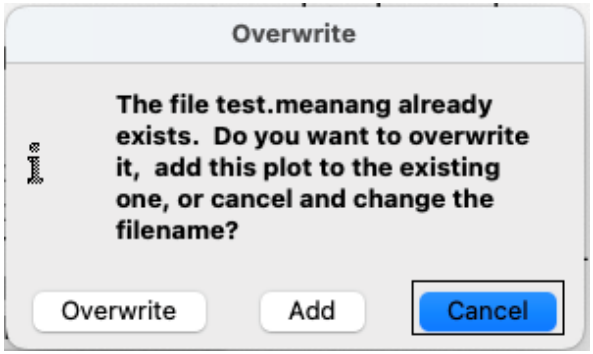
Include only secondary particles not originating in regions associated with “exclusive bits”

Input/Output files

Input phase space data

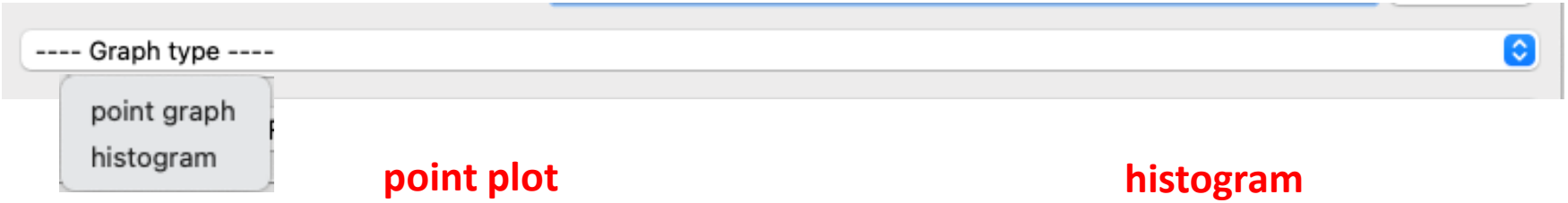


Output data file optimized for plotting with xmgrace/qtgrace

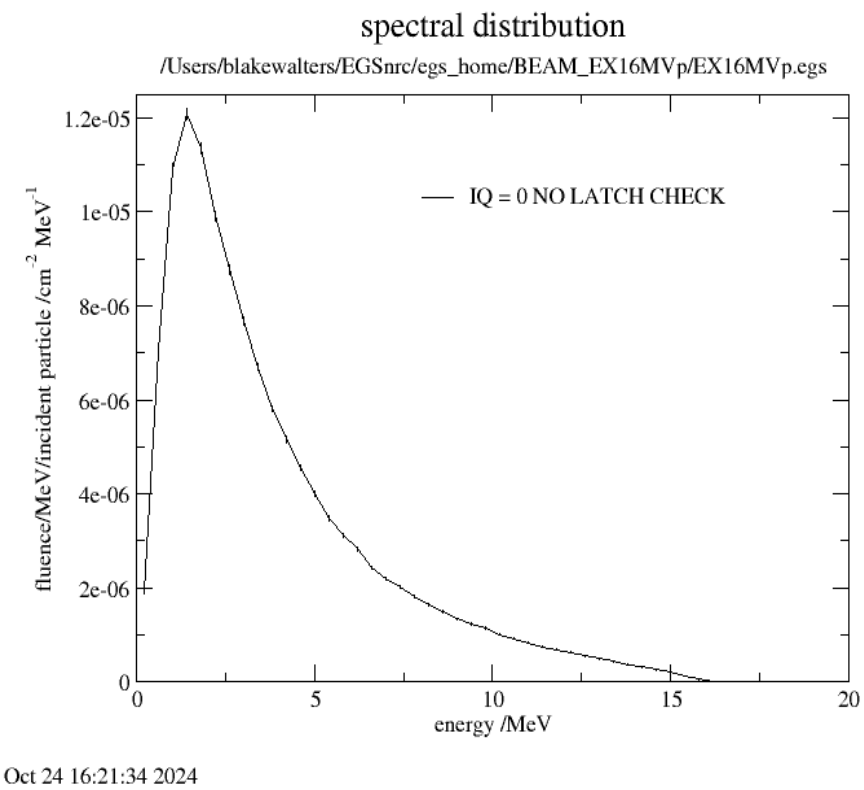


Use the same output file for the option to add data to an existing plot

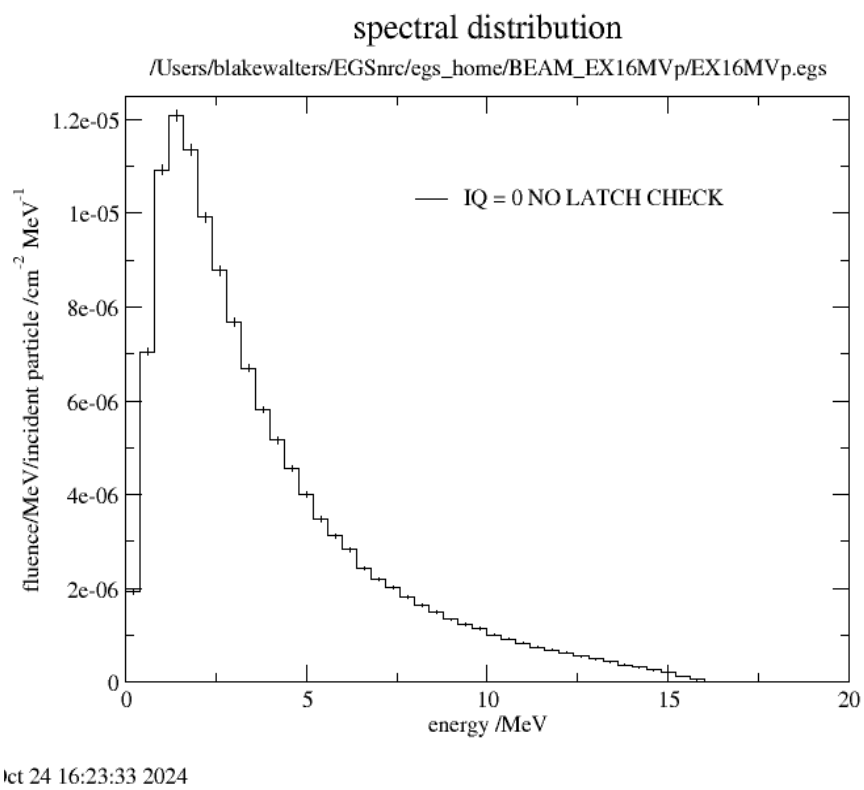
Plot type



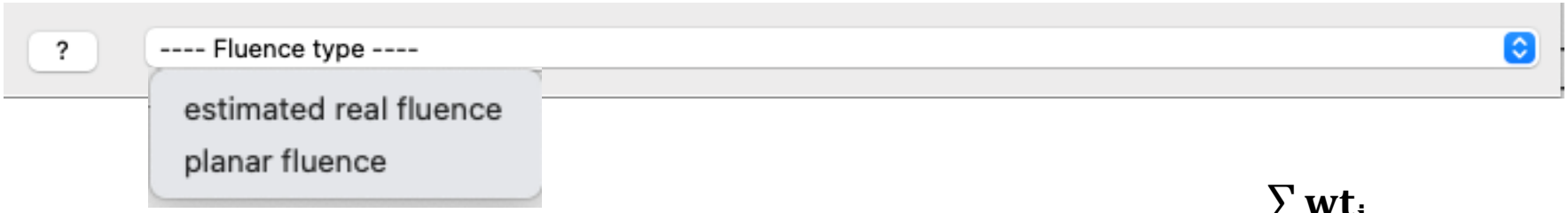
point plot



histogram



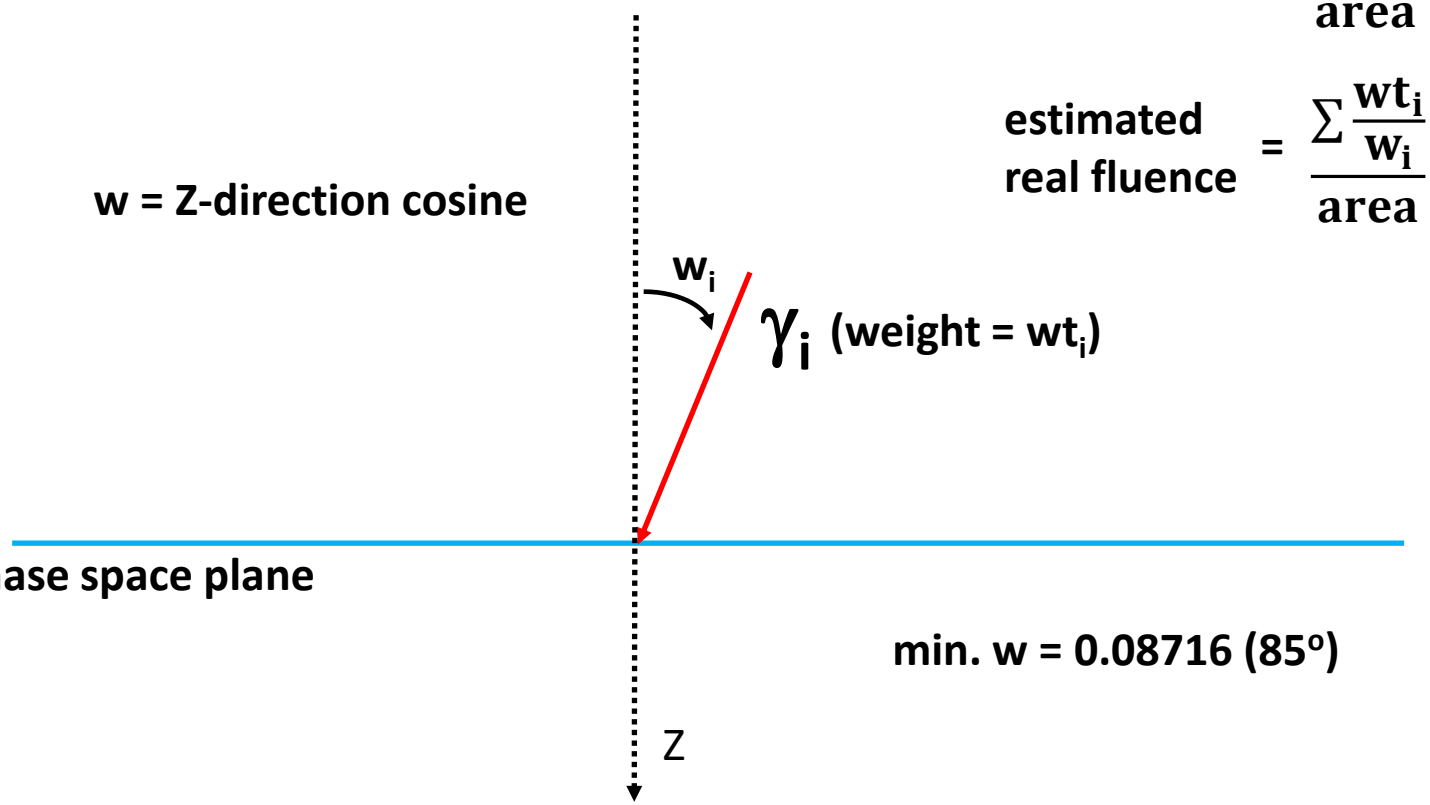
Fluence type



$$\text{planar fluence} = \frac{\sum \mathrm{wt_i}}{\mathrm{area}}$$

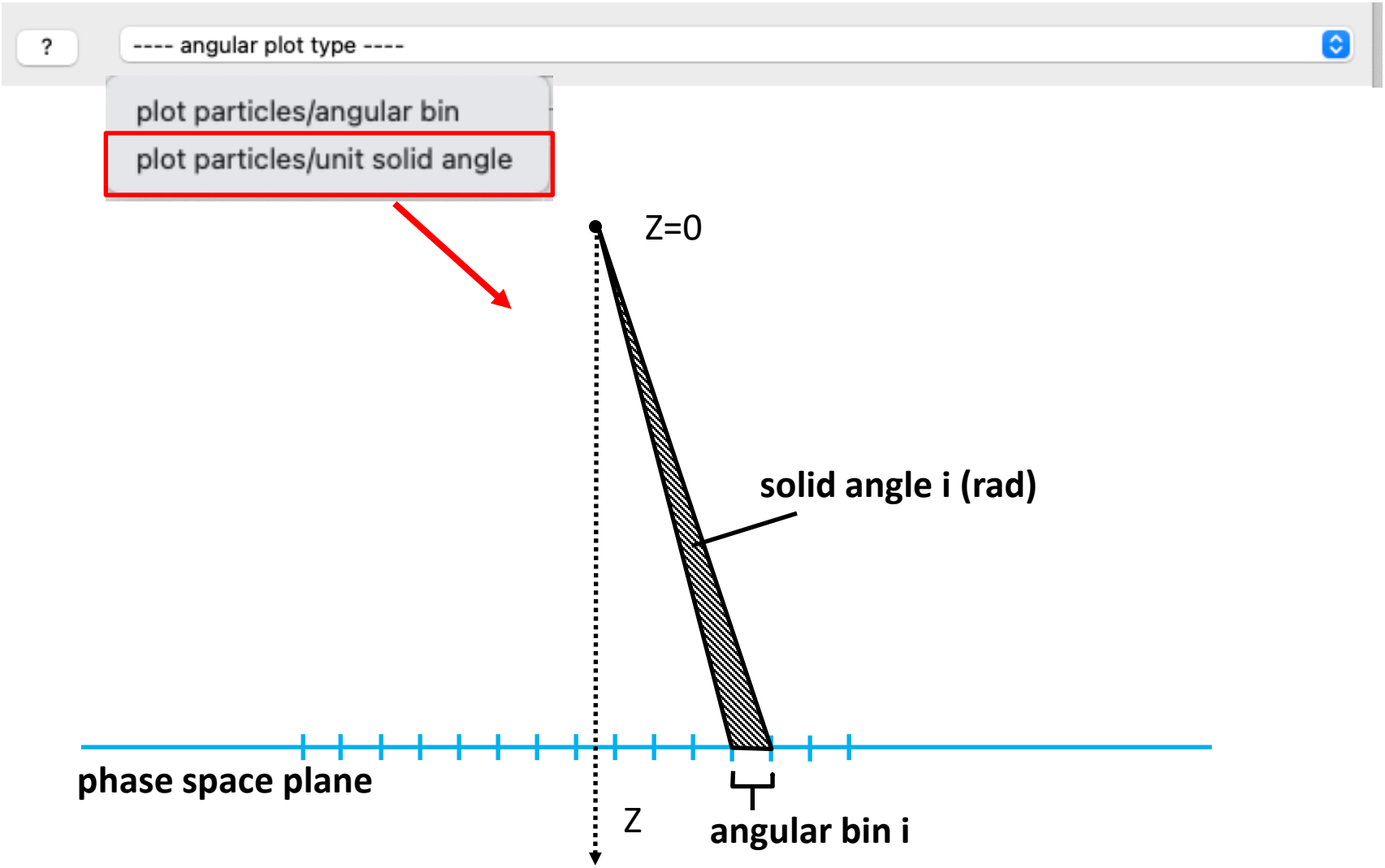
$$\text{estimated real fluence} = \frac{\sum \frac{\mathrm{wt_i}}{\mathrm{w_i}}}{\mathrm{area}}$$

w = Z-direction cosine

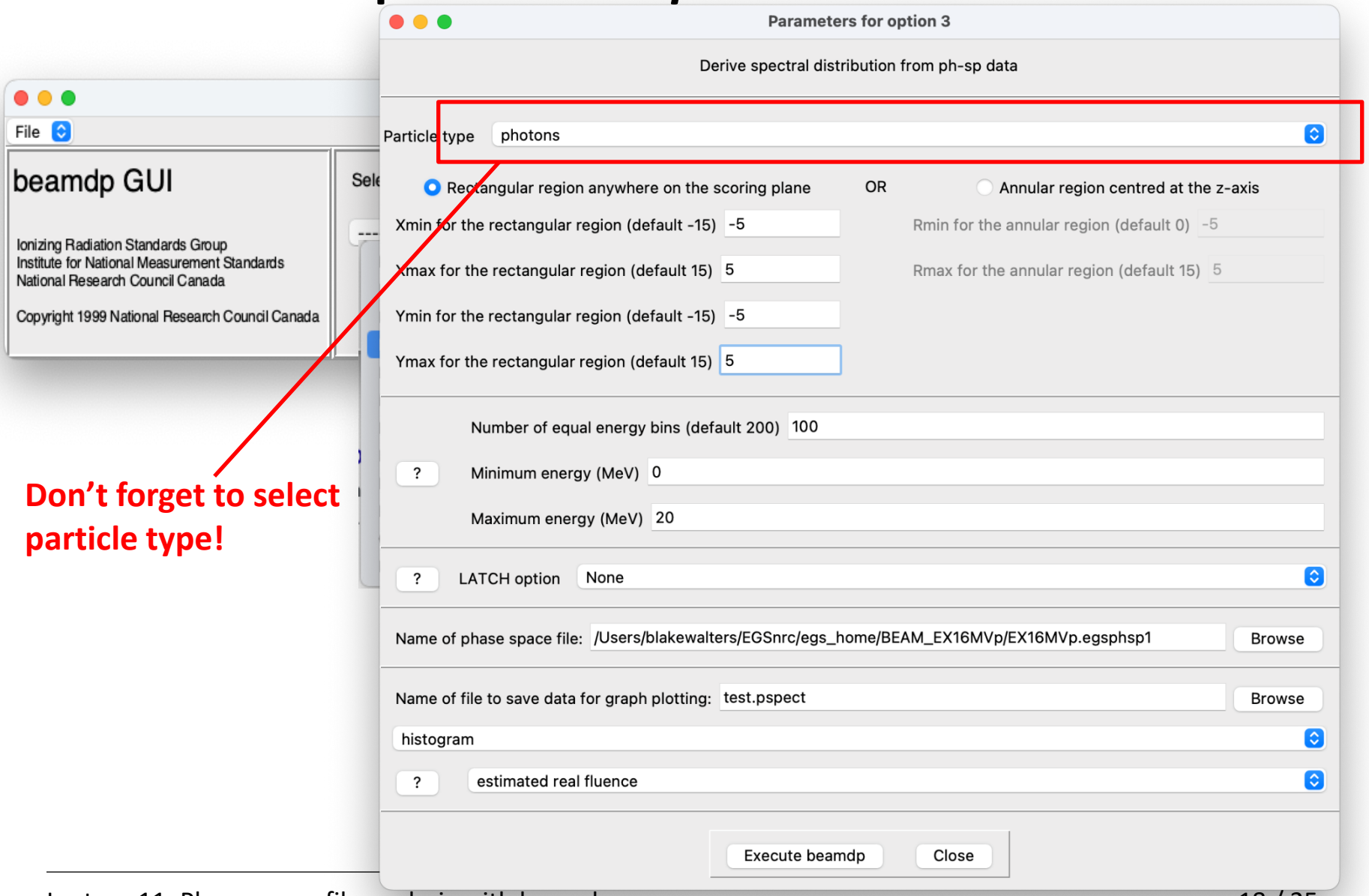


min. w = 0.08716 (85°)

Angular distributions



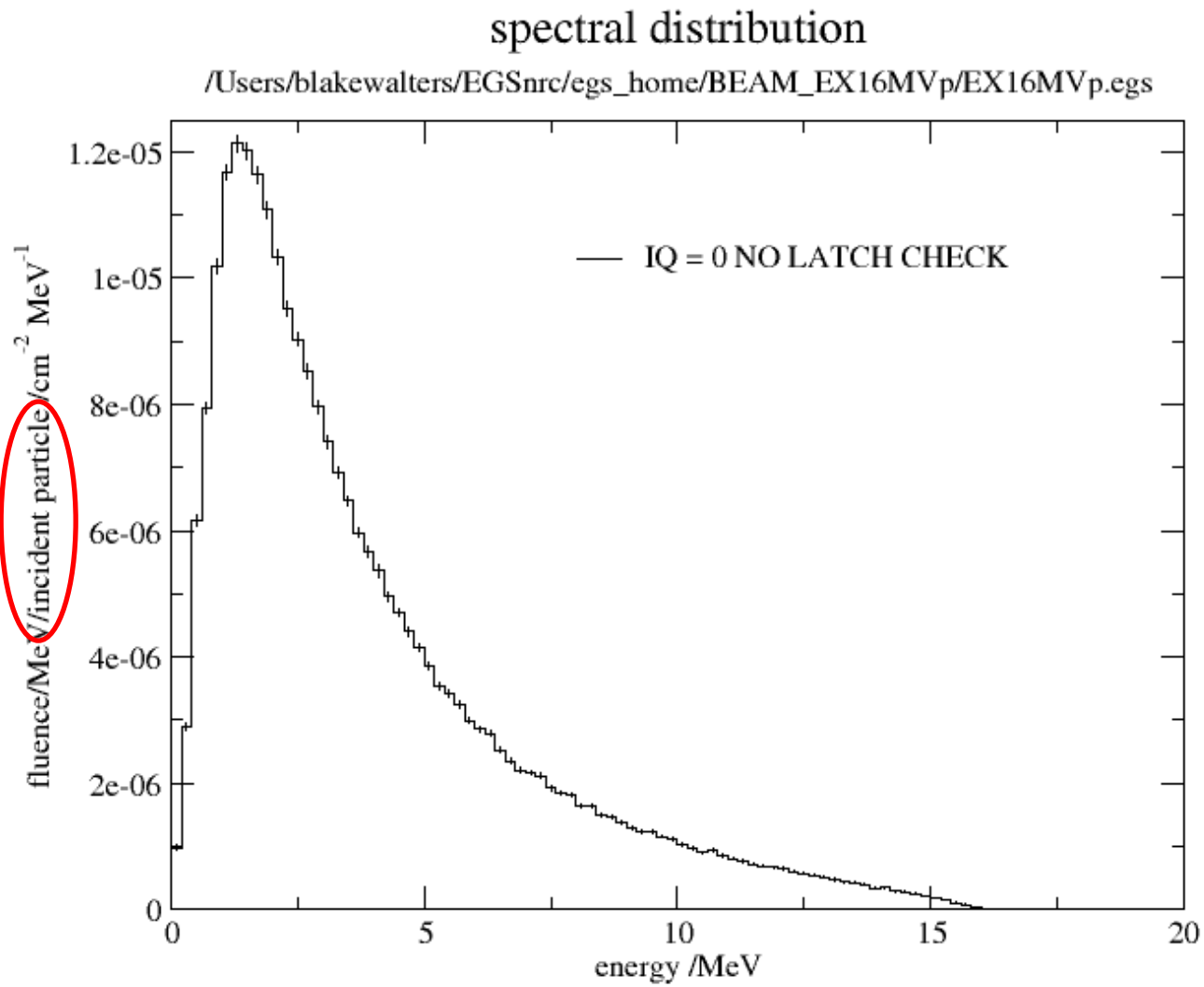
Example: *Derive spectral distribution*



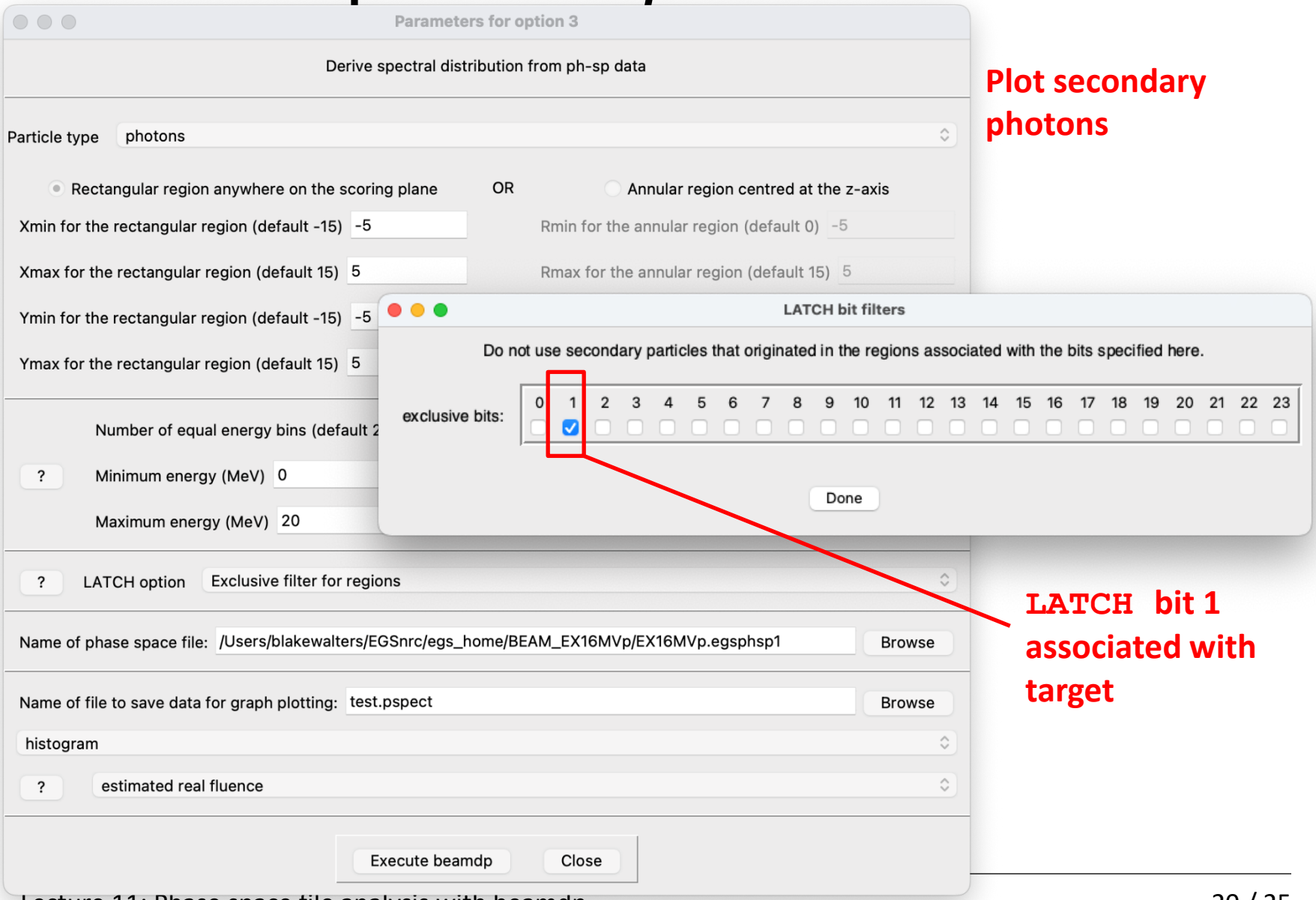
Don't forget to select particle type!

Example: *Derive spectral distribution*

statistically-independent histories (i.e. e- incident on target)



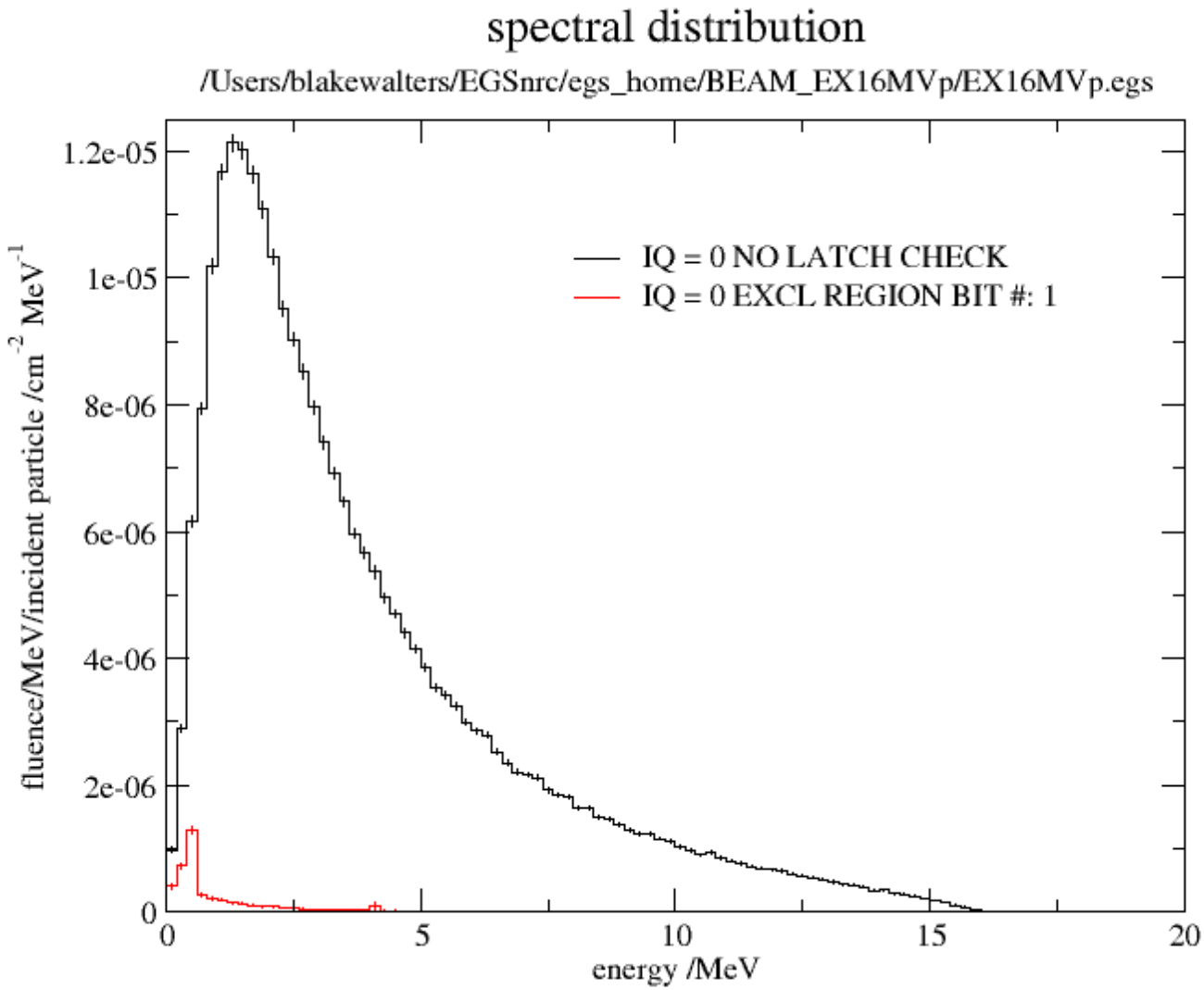
Example: *Derive spectral distribution*



Plot secondary photons

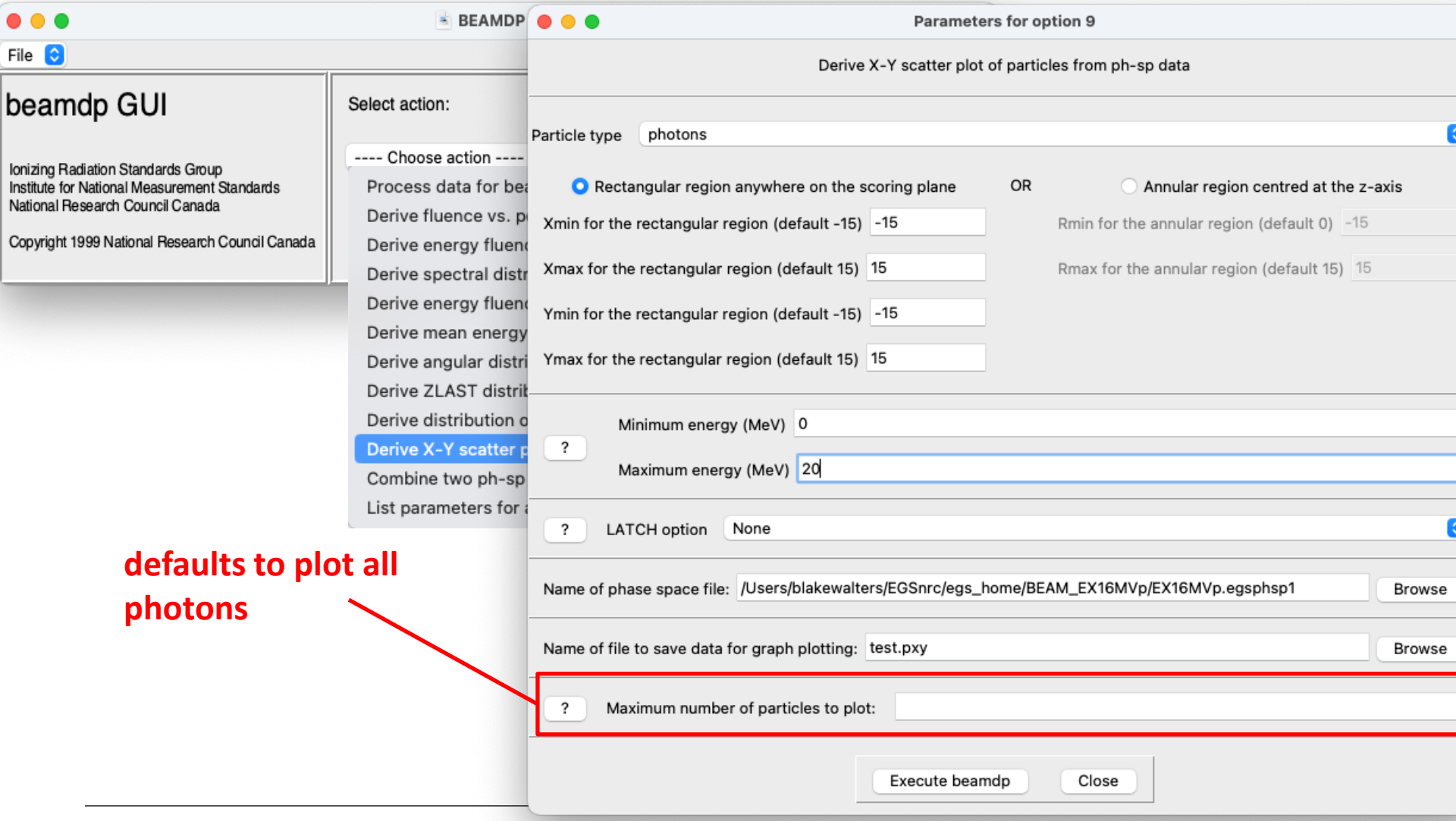
LATCH bit 1 associated with target

Example: *Derive spectral distribution*

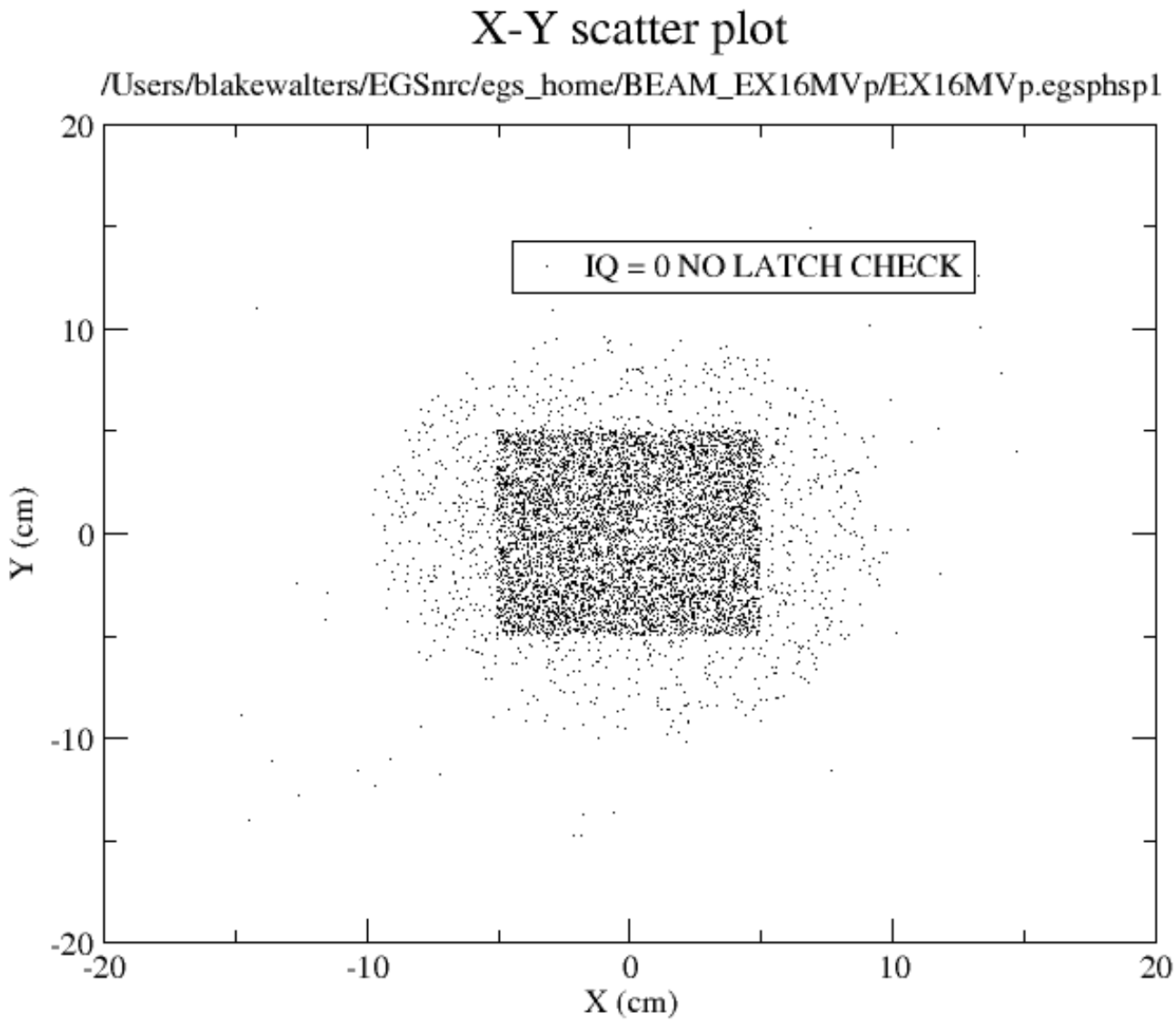


Thu Oct 24 16:32:56 2024

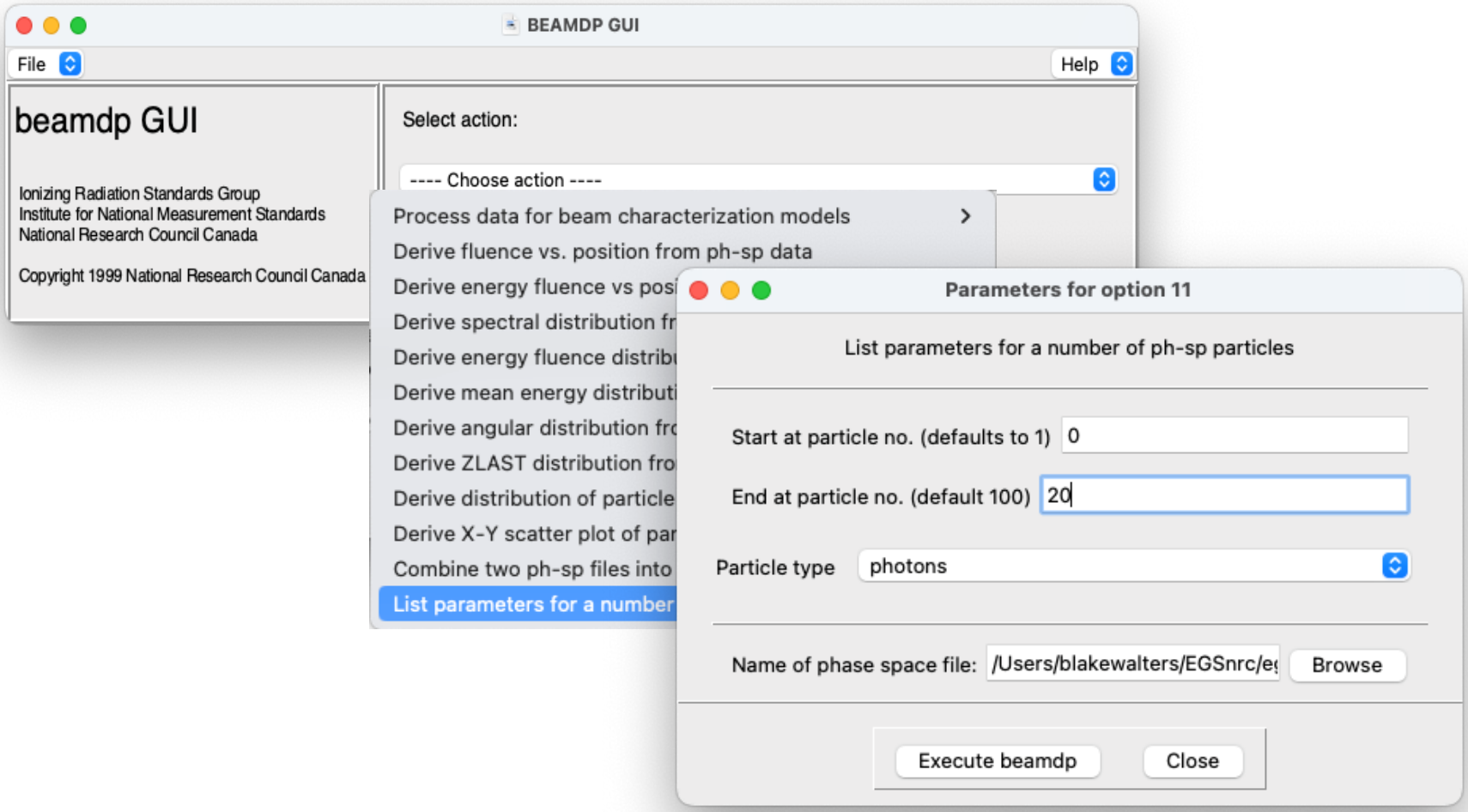
X-Y Scatter plot



defaults to plot all photons


X-Y scatter plot
/Users/blakewalters/EGSnrc/egs_home/BEAM_EX16MVp/EX16MVp.egsphsp1
IQ = 0 NO LATCH CHECK
Y (cm)
X (cm)
20
10
0
-10
-20
-20
-10
0
10
20

List parameters for a number of particles



List parameters for a number of particles

beamdp Output

```
IQ = 0 (only for photons)

Name of file containing phase space data (with ext., < A100):
 File name input is:/Users/blakewalters/EGSnrc/egs_home/BEAM_EX16MVp/EX16MVp.egsphsp1

First, try to open it as a MODE0 file

        Total number of particles in file:     5754377
              Total number of photons:     5300754
The rest are electrons/positrons.

    Maximum kinetic energy of the particles:     19.991 MeV
    Minimum kinetic energy of the electrons:      0.189 MeV
# of incident particles from original source:     50000.0

ENERGY  IQ    X      Y     U     V     W   WEIGHT      LATCH (set=1, not set=0)

 1.590   0   4.523   1.546  0.045  0.015  0.999 9.731E-04  00001 00000000000000000000011

ENERGY  IQ    X      Y     U     V     W   WEIGHT      LATCH (set=1, not set=0)

 1.997   0  -0.381  -1.565 -0.004 -0.016  1.000 9.754E-04  00001 00000000000000000000011
 7.322   0   1.157   0.019  0.012  0.000  1.000 9.767E-04  00001 00000000000000000000011
 0.257   0   0.837  -2.299  0.008 -0.023  1.000 8.057E-04  00001 00000000000000000000011
 0.689   0  -1.843  -2.906 -0.018 -0.029  0.999 9.515E-04  00001 00000000000000000000011
 0.181   0  -1.810  -0.141 -0.018 -0.001  1.000 6.187E-04  00001 00000000000000000000011
 0.368   0  -0.308  -2.003 -0.003 -0.020  1.000 8.958E-04  00001 00000000000000000000011
 1.156   0  -0.557  -4.960 -0.006 -0.050  0.999 9.678E-04  00001 00000000000000000000011
 2.462   0  -4.439  -2.943 -0.044 -0.029  0.999 9.767E-04  00001 00000000000000000000011
 0.759   0   0.416  -2.143  0.004 -0.021  1.000 9.555E-04  00001 00000000000000000000011
 0.263   0   4.185  -2.664  0.042 -0.027  0.999 8.139E-04  00001 00000000000000000000011
 1.202   0   0.441   1.495  0.004  0.015  1.000 9.687E-04  00001 00000000000000000000011
 1.193   0   3.102   3.625  0.070  0.067  0.995 3.132E-05  00001 10000000000000000000011
15.336   0  -0.609  -1.518 -0.006 -0.015  1.000 9.718E-04  00001 00000000000000000000011
 1.602   0   2.675  -1.463  0.027 -0.015  1.000 9.732E-04  00001 00000000000000000000011
 0.108   0   0.112  -0.475  0.001 -0.005  1.000 1.718E-04  00001 00000000000000000000011
 0.782   0   4.275  -4.788  0.043 -0.048  0.998 9.566E-04  00001 00000000000000000000011
13.642   0   0.104  -1.875  0.001 -0.019  1.000 9.727E-04  00001 00000000000000000000011
 1.865   0   0.138   4.953  0.002  0.120  0.993 2.726E-05  01100 00000000001000000000011
 0.336   0   0.798  -1.102  0.008 -0.011  1.000 8.798E-04  00001 00000000000000000000011

BYE!


If running gui, this window will close before next run!
```